

Disentangling Safety and Utility in LLM Activation Spaces

Team 9: Shrey Katyal, Vansh Amara, Amogh Paruvelli, Aditya Machiroutu, Rohan Abraham, Kushal

Motivation

- Aligned LLMs need to refuse harmful prompts and stay useful for normal ones
- Prior work shows both behaviors map to linear directions in activation space (Arditi 2024, Wei 2024)
- We don't know if safety and utility live in separate subspaces or share structure

Our Approach

- Extract safety and utility directions with two methods: Difference-of-Means and ActSVD
- Ablate each direction with a forward hook on a single transformer block
- Run a 2×2 cross-intervention matrix on Gemma-2-2B-IT & Llama-3.1-8B-Instruct
- If separable, ablating one direction breaks only its own behavior

Related Work

Prior Work

- Arditi et al. (2024): refusal in LLMs is mediated by a single direction recovered from a difference of means; we adopt their DoM method
- Wei et al. (2024): safety lives in low-rank subspaces found via SVD; we adopt their ActSVD method
- Zou et al. (2023): representation engineering. High-level behaviors map to readable directions
- Marks & Tegmark (2023): the difference-of-means technique formalized for truthfulness
- Mazeika et al. (2024): HarmBench. Safety often localizes to a small number of parameters

Our Contribution

- Most prior work studies safety in isolation
- We extract both safety AND utility on the same model
- We run all 4 ablation cells (DoM $\times$ ActSVD $\times$ safety $\times$ utility) plus baseline
- We use principal angles, not just cosine, to characterize subspace overlap
- We test on two models at different scales

Method

Direction Extraction

- **DoM (safety):** $r_s = \text{mean}(h_{\text{harmful}}) - \text{mean}(h_{\text{harmless}})$, normalized; AdvBench vs Alpaca, last prompt token, N=256
- **DoM (utility):** $r_u = \text{mean}(h_{\text{high}}) - \text{mean}(h_{\text{low}})$, normalized; HelpSteer pairs with $\delta \geq 3$, response tokens averaged, N=256
- **ActSVD:** take the SVD of $D = H_{\text{pos}} - H_{\text{neg}}$, keep top-k right singular vectors as subspace basis V (rank=4)

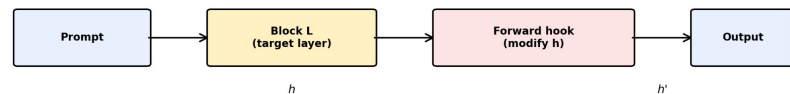
Forward-Hook Ablation (single transformer block)

- **DoM:** $h' = h - (h \cdot \hat{f})\hat{f}$ (subtract component along the direction)
- **ActSVD:** $h' = h - (hV^T)V$ (subtract projection onto the subspace)
- No model weights are changed; fully reversible

2x2 Cross-Intervention Matrix (5 cells per model)

- baseline (no ablation)
- dom_safety, dom_utility
- actsvd_safety, actsvd_utility
- Each cell scored on AdvBench ASR + BoolQ + HellaSwag + ARC-C + TruthfulQA (control)

Forward-Hook Ablation (single transformer block)



DoM: $h' = h - (h \cdot \hat{f})\hat{f}$ (subtract component along one direction)

ActSVD: $h' = h - (hV^T)V$ (subtract projection onto rank-4 subspace)

No model weights are changed. Hook is fully reversible.

Expected Pattern if Safety and Utility Are Separable

	Ablate safety	Ablate utility
DoM	ASR up Utility flat	ASR flat Utility down
ActSVD	ASR up Utility flat	ASR flat Utility down

Anything else means safety and utility are NOT cleanly separable.

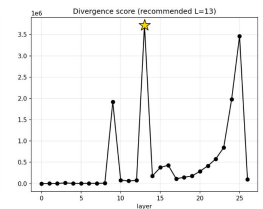
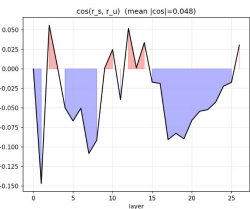
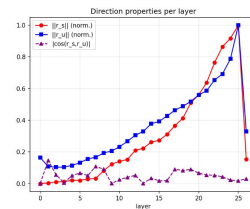
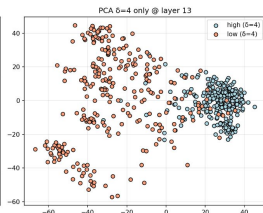
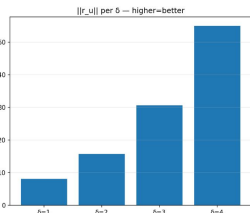
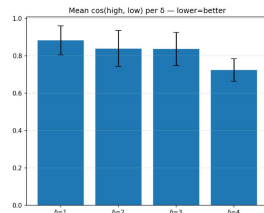
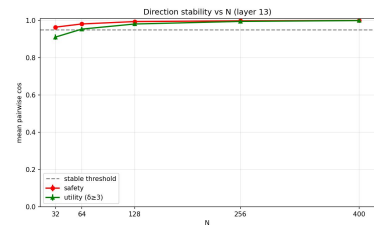
EDA + Layer Choice

EDA findings

- Layer-wise orthogonality: mean $|\cos(r_s, r_u)| = 0.048$ Gemma, 0.134 Llama
- δ threshold: utility direction norm scales monotonically; $\delta=4$ norm is 3-7 $\times$ the $\delta=1$ norm
- Bootstrap stability: pairwise cosine ≥ 0.99 at $N=256$ with $\delta \geq 3$
- Defaults justified: $N=256$, $\delta \geq 3$, rank=4

Layer choice

- Gemma: EDA peaks at index 26; we sweep {22, 23, 24, 25} and pick $L=25$ (where ablation actually works)
- Llama: EDA peaks at index 32, past the last hookable transformer block. $L=31$ had no effect. Swept 13-17 and 18-28, picked $L=16$ as best available
- $L=25$ norm spike originally flagged as artifact is in fact the most effective layer



Cross-Intervention Results: Gemma-2-2B-IT

Three of four cells show clean separation

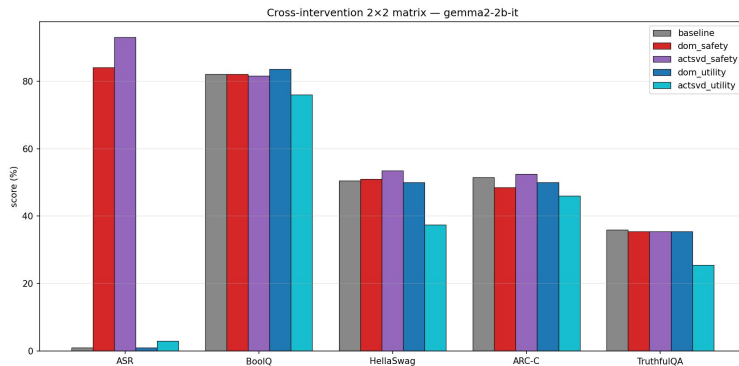
- dom_safety: ASR jumps from 1% to 84%; utility flat. r_s is causally necessary for refusal at L=25
- dom_utility: nothing moves; the DoM utility direction does not gate either capability
- actsvd_safety: even stronger. ASR 93%, utility actually nudges UP
- actsvd_utility: BREAKS the pattern. Utility drops 8 points, TruthfulQA drops 10 points

Why does actsvd_utility break? (geometric mechanism)

- Principal angles V_s vs V_u : 61.9° , 80.7° , 83.3° , 89.9°
- Three angles are nearly orthogonal; one sits at $\sim 62^\circ$
- That one component shares $\sim 22\%$ of its variance with the safety subspace
- Ablating V_u removes some safety structure as a side effect
- Too small to flip refusal, big enough to break utility computations

Results Table

Condition	ASR	Avg Utility	TruthfulQA
baseline	0.01	0.613	0.360
dom_safety	0.84	0.605	0.355
dom_utility	0.01	0.612	0.355
actsvd_safety	0.93	0.625	0.355
actsvd_utility	0.03	0.532	0.255



Cross-Intervention Results: Llama-3.1-8B-Instruct

What this shows

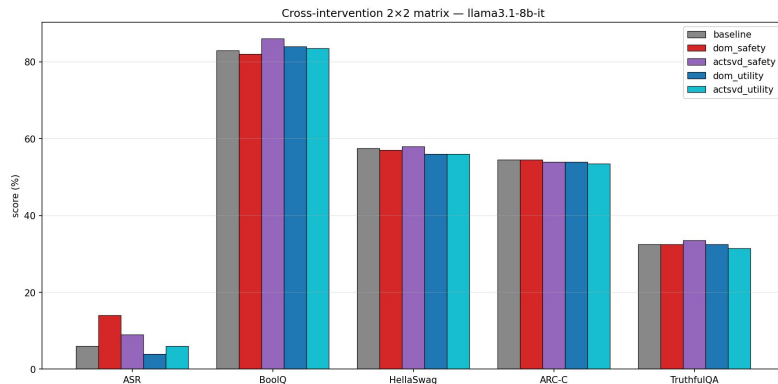
- No ablation cell produces a strong effect; max ASR movement is 6% $\rightarrow$ 14%
- Utility essentially flat across all 5 cells
- Geometry is nearly identical to Gemma: $\phi = 0.074$ vs 0.065; principal angles 61.0°, 79.3°, 81.2°, 87.8°
- So geometric near-orthogonality alone does NOT predict functional independence
- Possible reasons: safety spread across layers, redundant directions, chat-template / refusal-style differences

Sensitivity Sweeps (Gemma)

- Layer sweep at dom_safety: ASR 0.04 (L=22) $\rightarrow$ 0.41 (L=23) $\rightarrow$ 0.55 (L=24) $\rightarrow$ 0.84 (L=25); one layer off and the effect collapses
- Rank sweep at actsvd_safety: rank 1 $\rightarrow$ 0.86, rank 4 $\rightarrow$ 0.93, rank 8 $\rightarrow$ 0.97
- Above rank 4, utility starts to slide (0.62 $\rightarrow$ 0.58 $\rightarrow$ 0.55); rank 4 is the sweet spot

Results Table

Condition	ASR	Avg Utility
baseline	0.06	0.650
dom_safety	0.14	0.645
dom_utility	0.04	0.647
actsvd_safety	0.09	0.660
actsvd_utility	0.06	0.643



Conclusions

Findings

- Gemma: 3 of 4 cells support clean separation; 1 cell entangled, traced to a single shared subspace component at $\sim 62^\circ$
- Llama: nearly identical geometry, but no strong ablation effect. Reported as negative finding
- Geometric near-orthogonality is necessary but not sufficient for functional independence
- ActSVD is more aggressive than DoM at breaking safety, with no utility cost (Gemma)

Limitations

- ASR uses substring matching, not an LLM judge
- Two-model evidence; Llama result could be model-specific
- Single-layer ablation only
- HelpSteer helpfulness is one definition of utility

Future Work

- LLM-judge ASR rescoring of cached generations
- Mechanistic look at the $\sim 62^\circ$ shared subspace component
- Multi-layer interventions for the Llama case

Thank You!

Code & results: github.com/shreyk2/nlp-team-9